The Causal Hierarchy: Why General Relativity Operates Where Quantum Field Theory Cannot

A Critical Assessment

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Abstract

We present a precise formulation of the following observation: in the interior of the Schwarzschild black hole, there exists a regime arbitrarily close to the singularity where the spacetime metric and curvature are well-defined quantities of classical general relativity, but no quantum field process—no particle interaction, no gauge boson exchange, no dynamical cycle of any kind—can complete. This follows from two established facts: (i) general relativity describes gravity not as a force but as the curvature of spacetime, encoded in the metric tensor at each point, and (ii) all fundamental forces of nature are mediated by quantum field processes that require finite temporal extent. The resulting hierarchy—geometry precedes dynamics, and dynamics requires causal structure—is derived entirely from standard physics. No modification of either general relativity or quantum mechanics is proposed. We define *nothingness* as the limit in which the maximal causal diamond shrinks to zero four-volume, and we discuss the implications of the hierarchy for the relationship between gravity and quantum theory.

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1 Introduction

The reconciliation of general relativity and quantum mechanics is widely regarded as the central open problem of fundamental physics [1, 2]. The dominant research programs—string theory [3], loop quantum gravity [4], and asymptotic safety [5]—share a common assumption: that gravity must ultimately be described by a quantum theory.

This assumption deserves scrutiny, and the Schwarzschild black hole provides a concrete arena in which to examine it. As Dyson observed [6], the correct direction of unification may be understanding how quantum phenomena require the gravitational arena, rather than quantizing gravity itself.

We make no speculative claims. The content of this paper follows from two premises that are not in dispute:

1. General relativity describes gravity as the curvature of spacetime. The metric tensor $g_{\mu\nu}$ and the Riemann curvature tensor $R^{\alpha}{}_{\beta\gamma\delta}$ are defined at each point of the manifold. Gravity is not a force mediated by particle exchange; it is the geometry of the background.

2. All fundamental forces of nature—electromagnetic, strong, and weak—are described by quantum field theories. The interactions they mediate (virtual photon exchange, gluon exchange, W/Z exchange) are quantum processes that require finite proper time to complete.

From these two premises and the known causal structure of the Schwarzschild interior, a hierarchy follows.

2 The Schwarzschild Geometry

The Schwarzschild line element is

$$ds^2 = -f(r)\, c^2\, dt^2 + f(r)^{-1}\, dr^2 + r^2\, d\Omega^2\,, \tag{1}$$

where $f(r) = 1 - r_s/r$, $r_s = 2GM/c^2$, and $d\Omega^2 = d\theta^2 + \sin^2\theta\, d\phi^2$.

In the exterior ($r > r_s$), the coordinate t is timelike and r is spacelike. In the interior ($0 < r < r_s$), $f < 0$: the coordinate r becomes timelike and the singularity at $r = 0$ lies in every observer's causal future. The horizon at $r = r_s$ is a null surface where the coordinate singularity is removable [8]; the geometry is smooth there.

The Kretschner scalar, which measures the magnitude of spacetime curvature independently of coordinates, is

$$K = R_{\alpha\beta\gamma\delta}\, R^{\alpha\beta\gamma\delta} = \frac{48\, G^2 M^2}{c^4\, r^6}\,. \tag{2}$$

This is finite for all $r > 0$ and diverges only at $r = 0$.

3 The Finite Causal Future of the Interior

Proposition 1 (Bounded proper time to the singularity)**.** *Let p be an event in the Schwarzschild interior at areal radius $R_0 < r_s$. The maximum proper time from p to*

the singularity $R = 0$, along any future-directed causal curve, is

$$\tau_{\rm coh}(R_0) = \frac{r_s}{c}\left[\arccos\sqrt{\frac{R_0}{r_s}} + \sqrt{\frac{R_0}{r_s}\left(1 - \frac{R_0}{r_s}\right)}\,\right], \tag{3}$$

with $\tau_{\rm coh}(r_s) = \pi r_s/(2c)$ and $\tau_{\rm coh}(0) = 0$.

Proof. The interior contains a trapped surface. By the Penrose singularity theorem [9], all future-directed causal curves from p reach the singularity in finite affine parameter. For a radial free-fall geodesic, the equation of motion gives [10]

$$\left(\frac{dR}{d\tau}\right)^2 = \frac{r_s\, c^2}{R}\,. \tag{4}$$

Since R decreases toward the future, $dR/d\tau = -c\sqrt{r_s/R}$. Separating variables and integrating from R_0 to 0 with the substitution $R = r_s \cos^2\eta$:

$$\tau_{\rm coh} = \frac{1}{c}\int_0^{R_0}\sqrt{\frac{R}{r_s}}\,dR = \frac{r_s}{c}\int_0^{\eta_0} 2\cos^2\eta\,d\eta = \frac{r_s}{c}\left[\eta_0 + \frac{1}{2}\sin 2\eta_0\right], \tag{5}$$

where $\cos\eta_0 = \sqrt{R_0/r_s}$, yielding (3). □

Remark 1. *In the exterior ($R > r_s$), future-directed timelike curves can be extended to infinite proper time. The causal future of any exterior event is infinite. In Minkowski space, the causal future is likewise infinite at every event. The finite causal future is a feature unique to the interior.*

4 The Nature of Gravity versus the Nature of Forces

This section contains no new results. It restates established physics in a form that makes the subsequent argument explicit.

4.1 Gravity is geometry, not a force

In general relativity, the gravitational "interaction" is not mediated by particle exchange. A freely falling body follows a geodesic of the spacetime metric:

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\alpha\beta}\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau} = 0\,. \tag{6}$$

The metric $g_{\mu\nu}(x)$, the connection $\Gamma^\mu_{\alpha\beta}(x)$, and the curvature $R^\alpha{}_{\beta\gamma\delta}(x)$ are all defined *at each point* of the manifold. Their definition requires no temporal extent, no dynamical process, and no exchange of quanta. They are properties of the arena, not of processes occurring within it.

This remains true in the Schwarzschild interior: for every $r > 0$, the metric, connection, and curvature are well-defined and finite. The Einstein vacuum equation $R_{\mu\nu} = 0$ is satisfied everywhere except at $r = 0$.

4.2 All fundamental forces are quantum field processes

In the Standard Model of particle physics [13]:

- The electromagnetic interaction is described by quantum electrodynamics (QED). Charged particles interact through the exchange of virtual photons.
- The strong interaction is described by quantum chromodynamics (QCD). Quarks interact through gluon exchange, and nucleon binding involves meson exchange processes.
- The weak interaction is mediated by $W^{\pm}$ and Z^0 bosons.

Each of these processes requires a finite interval of proper time. A virtual photon mediating the Coulomb interaction between an electron and a proton is not instantaneous; it propagates within the light cone and requires temporal extent determined by the energy-time uncertainty relation and the relevant coupling constants.

4.3 Particle identity is quantum mechanical

The properties that define a particle are quantum field-theoretic in origin:

- **Mass:** For fermions in the Standard Model, mass arises from the Yukawa coupling to the Higgs field—a quantum field process [12].
- **Charge:** Electric charge is defined through QED coupling. It is meaningless without the electromagnetic quantum field.
- **Particle identity:** An electron is an excitation of the electron field ψ_e. Its identity as a particle is a quantum field concept.
- **Bound states:** A hydrogen atom exists because the electron and proton exchange virtual photons periodically. A proton exists because quarks exchange gluons. These are sustained quantum field processes.

There is no classical, non-quantum description of any fundamental force or any particle's intrinsic properties at the fundamental level.

5 The Hierarchy

Proposition 2 (Existence of a regime where GR operates and QFT cannot)**.** *For any quantum field process with characteristic proper-time scale $T > 0$ (for example, the Compton time $T = \hbar/(mc^2)$ associated with a particle of mass m), there exists a radius $R_T > 0$ in the Schwarzschild interior such that:*

(a) For all R with $0 < R < R_T$: $\tau_{\text{coh}}(R) < T$. The process cannot complete even one cycle.

(b) For all R with $0 < R < R_T$: the metric $g_{\mu\nu}$, the connection $\Gamma^{\mu}_{\alpha\beta}$, and the Riemann tensor $R^{\alpha}{}_{\beta\gamma\delta}$ are finite and well-defined. The Einstein equation $R_{\mu\nu} = 0$ holds.

Proof. Statement (a): $\tau_{\text{coh}}(R)$ is continuous, monotonically increasing for $R \in (0, r_s)$, with $\tau_{\text{coh}}(0) = 0$. For any $T > 0$, by continuity, there exists $R_T > 0$ such that $\tau_{\text{coh}}(R) < T$ for all $R < R_T$.

Statement (b): The Kretschner scalar (2) is finite for all $r > 0$. The metric components in regular coordinates (e.g., Kruskal–Szekeres) are smooth for $r > 0$. The vacuum Einstein equation is satisfied identically for the Schwarzschild solution at all $r > 0$. These statements hold independently of the value of τ_{coh}. □

Corollary 1 (Progressive failure of quantum processes)**.** *As $R \to 0$:*

(i) The slowest quantum processes (e.g., nuclear beta decay, $T \sim 10^3$ s) become unsustainable first, at relatively large R.

(ii) Faster processes (atomic transitions, $T \sim 10^{-15}$ s) persist to smaller R.

(iii) The fastest known processes (involving the heaviest particles, $T \sim \hbar/(m_t c^2) \sim 10^{-25}$ s for the top quark) persist to yet smaller R.

(iv) For any $T > 0$, no matter how small, there exists a regime where $\tau_{\text{coh}} < T$ but the geometry is well-defined.

The geometry of spacetime is the last structure standing.

This establishes the hierarchy:

$$\text{Gravity (geometry)} \longrightarrow \text{Causal structure} \longrightarrow \text{Quantum field theory} \longrightarrow \text{Forces, particles, matter.} \tag{7}$$

General relativity provides the metric; the metric determines the causal structure (light cones, available proper time); the causal structure determines whether quantum field processes have sufficient temporal extent to operate; where they operate, quantum field theory gives rise to forces, particles, and bound states.

6 Nothingness

The singular point $r = 0$ is not part of the spacetime manifold; it is a boundary where the geometry itself ceases to be defined. Any physically meaningful definition of nothingness must therefore refer not to that mathematical limit but to a genuine region of spacetime—a region with finite, nonzero four-volume—where the causal diamond is too small for quantum field theory to operate.

Definition 1 (Process-relative nothingness)**.** *Let $\mathcal{P}$ be a quantum field process with characteristic proper-time scale $T_{\mathcal{P}} > 0$ (e.g., the Compton time $\hbar/(mc^2)$ of a particle of mass m, or the period of a gauge boson exchange cycle sustaining a bound state). The* nothingness region *of the Schwarzschild interior with respect to $\mathcal{P}$ is the set*

$$\mathcal{N}_{\mathcal{P}} = \left\{ p \in \mathcal{M} \;\middle|\; \tau_{\text{coh}}(R(p)) < T_{\mathcal{P}} \right\}. \tag{8}$$

This is an open region of the spacetime manifold with finite, nonzero four-volume.

Within $\mathcal{N}_{\mathcal{P}}$, the process $\mathcal{P}$ cannot complete even one cycle. The causal diamond available at each event is a genuine, finite region of curved spacetime—not a point, not a limit—but its temporal depth is insufficient for $\mathcal{P}$ to occur. The geometry (metric, curvature, geodesics, Einstein equations) is well-defined throughout $\mathcal{N}_{\mathcal{P}}$, since the Kretschner scalar (2) is finite for all $r > 0$.

Definition 2 (Nothingness)**.** *The* nothingness region *of the Schwarzschild interior is the set*

$$\mathcal{N} = \left\{ p \in \mathcal{M} \;\middle|\; \tau_{\text{coh}}(R(p)) < T_{\min} \right\}, \tag{9}$$

where $T_{\min}$ is the characteristic timescale of the fastest quantum field process that can be physically realised. A concrete upper bound is the Compton time of the heaviest known elementary particle (the top quark, $m_t \approx 173\ GeV/c^2$):

$$T_{\min} \leq \frac{\hbar}{m_t c^2} \approx 3.8 \times 10^{-27}\ s. \tag{10}$$

In the nothingness region $\mathcal{N}$, *no* quantum field process of any known kind can complete. No force can act, since every force is mediated by quantum field processes requiring $T \geq T_{\min}$. No particle can maintain its identity, since mass, charge, and species are conferred by quantum field interactions (Yukawa couplings, gauge couplings) that require temporal extent. No bound state can persist, since binding requires at least one cycle of gauge boson exchange.

The manifold, however, continues. The metric is smooth, curvature is finite (though large), geodesics are well-defined, and the Einstein vacuum equation $R_{\mu\nu} = 0$ holds. Spacetime geometry exists throughout $\mathcal{N}$, but physics—in the sense of forces, particles, and interactions—does not.

Nothingness is therefore not an abstraction or a limiting concept. It is a concrete, finite, open region of the spacetime manifold where the causal structure is too shallow for quantum field theory to operate, while the geometric structure of general relativity remains intact.

Remark 2 (Progressive onset)**.** *The nothingness regions are nested: $\mathcal{N}_{\mathcal{P}} \subset \mathcal{N}_{\mathcal{Q}}$ whenever $T_{\mathcal{P}} < T_{\mathcal{Q}}$. As an observer falls toward the singularity, progressively faster quantum processes become unsustainable. First, slow processes (nuclear beta decay, $T \sim 10^3$ s) fail at large R. Then atomic processes ($T \sim 10^{-15}$ s), then hadronic processes ($T \sim 10^{-24}$ s). The nothingness region $\mathcal{N}$ is the innermost: the region where* nothing *quantum survives. But crucially, $\mathcal{N}$ has nonzero four-volume and is a genuine part of the spacetime.*

Remark 3 (Distinction from the quantum vacuum)**.** *This notion of nothingness is fundamentally different from the "nothing" discussed by Krauss [14] and others in the context of quantum cosmology. The quantum vacuum is a state* within *quantum field theory—the lowest-energy state of a quantum field on a fixed background. It presupposes that QFT is defined. Nothingness, as identified here, is a region of spacetime where QFT itself lacks sufficient causal future to operate. It is prior to the quantum vacuum, not a special case of it.*

Remark 4 (The exterior)**.** *In the exterior ($R > r_s$) and in Minkowski space, $\tau_{\text{coh}} = \infty$. No process, however slow, is excluded by the causal structure. The nothingness regions are empty: $\mathcal{N}_{\mathcal{P}} = \varnothing$ for all $\mathcal{P}$. This is why stable atoms, permanent bound states, and indefinite quantum coherence are possible in the exterior but not in the deep interior.*

7 The Equivalence Principle and Local Validity

The Einstein equivalence principle [10] guarantees that in a sufficiently small neighbourhood of any event—including events inside the horizon—the metric can be approximated by the Minkowski metric, and local physics is indistinguishable from special relativity.

We affirm this without reservation. A freely falling observer inside the horizon of a large black hole can conduct quantum experiments, observe standard particle interactions, and verify local QFT—*provided* the experiment's duration is much smaller than $\tau_{\text{coh}}(R)$ at the observer's location and the spatial extent is small compared to the curvature scale.

For a supermassive black hole ($M \sim 10^9\, M_\odot$), τ_{coh} at the horizon is approximately 5 hours. Local quantum experiments with duration much less than 5 hours proceed normally. There is no instantaneous violation of quantum mechanics at horizon crossing.

The hierarchy identified in this paper is a *global* statement about which structures can be sustained indefinitely, not a local statement about what physics applies at a given event. Locally, QFT is always valid (where curvature is sub-Planckian and τ_{coh} exceeds the experiment's duration). Globally, only geometry persists all the way to $R \to 0^+$.

8 Discussion

8.1 What this argument does not assume

We have not assumed that gravity is classical. The hierarchy follows regardless of whether gravity is ultimately quantum. Even if a quantum theory of gravity exists, the structural observation remains: the metric is defined pointwise, while forces require temporal extent. Whether the metric itself has quantum fluctuations is a separate question that this argument does not address.

However, the hierarchy is *consistent with* gravity being fundamentally classical, and it provides a concrete physical scenario—the Schwarzschild interior—where classical geometry operates in a domain where quantum field theory cannot sustain its processes. This constitutes evidence (not proof) that the assumption of quantum gravity is not inevitable.

8.2 What this argument does not claim

We do not claim that quantum mechanics is "wrong" or "breaks down" in any local sense inside the horizon. Local QFT validity is guaranteed by the equivalence principle and we affirm it.

We do not propose any new fields, symmetries, or interactions. No "causal Higgs mechanism" or "mass creation" is asserted. The hierarchy stands on established physics alone.

We do not claim to have resolved the black hole information paradox, explained dark matter, or constructed a theory of everything. The hierarchy is an observation about the logical structure of existing theories, not a new theory.

8.3 On the quantization of gravity

The hierarchy suggests—but does not prove—a reason for caution about quantizing gravity. The suggestion is:

1. Quantum mechanics requires causal structure to define coherent states and unitary evolution.
2. Causal structure is determined by the metric, which is the dynamical variable of general relativity.
3. Quantizing the metric means placing the causal structure in superposition.
4. A superposition of causal structures is a superposition of the conditions under which superposition itself is defined.

This is a conceptual tension, not a no-go theorem. It may be that a consistent formulation exists that resolves this circularity. But the Schwarzschild interior provides an existence proof of a regime where geometry persists and quantum processes do not, which is at minimum consistent with the possibility that gravity need not be quantized.

8.4 Limitations

The argument is presented for the Schwarzschild spacetime, which is a vacuum solution with exact spherical symmetry. Extension to Kerr, charged, or dynamical spacetimes would strengthen the case. The Penrose singularity theorem guarantees finite causal futures in the interior of any black hole with a trapped surface, so the qualitative conclusion likely extends, but the explicit formula (3) would need to be replaced by appropriate bounds.

We have also been silent on what happens *at* $r = 0$. There, curvature diverges, the manifold terminates, and neither GR nor QFT is well-defined. The hierarchy concerns the approach to the singularity ($r \to 0^+$), not the singular point itself.

9 Conclusion

The Schwarzschild interior reveals a structural asymmetry between general relativity and quantum field theory. General relativity describes gravity as the curvature of spacetime—a property defined at each point of the manifold, requiring no temporal extent. Quantum field theory describes all other forces as processes—gauge boson exchanges, field interactions—that require finite proper time to occur. Near the singularity, the available proper time shrinks to zero while the geometry remains well-defined.

This means there exist regions of spacetime where the arena (geometry) is present but the actors (quantum fields, forces, particles, bound states) cannot perform. The hierarchy

$$\text{Geometry} \longrightarrow \text{Causal structure} \longrightarrow \text{Quantum field theory}$$

is not a postulate. It is a consequence of what general relativity and quantum field theory already are, applied to a spacetime that already exists in the theory.

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